

1 ☆ QUESTION: 1 GROUP: Aptitude SECTION: Data Analysis Mark(s): 1

2 **Direction:** Study the following information and answer the question that follows:

3 The line graph shows the number of students (male and female included) enrolled in singing and dance class from 2000 to 2003.



9 The table shows the number of students (male and female included) enrolled in the dance class from different age groups in different years.

10

Years	Age groups		
	10-17 years	18-25 years	26-33 years
2000	46	60	55
2001	40	65	70
2002	55	60	64
2003	54	70	75

11

12

13

14 **Question:** What is the average number of students in the dance class whose age were either less than 10 years or more than 33 years in the given years?

Previous Question

Next Question

1

2

There are 4 mathematical expressions given below. How many are correct?

3

$$867898786898 \neq 867898786898$$

$$22844422444848 > 22844224444848$$

4 ☆

$$33553535353 < 33553353535$$

$$454555455545 = 454555455545$$

5

6

☐ 2

7

☐ 0

8

☐ 1

9

☐ 3

10

11

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QUESTION 5 GROUP: Aptitude SECTION: Quant Math Ability Mark(s): 1

Let α, β be the roots of the quadratic equation $2x^2 - 5x + 1 = 0$. If $S_n = (\alpha)^{2n} + (\beta)^{2n}$ then find the value of $(4S_{2021} + S_{2019}) / (S_{2020})$

☐ 26☐ 21☐ 25☐ 24

Previous Question

Next Question

QUESTION 7 GROUP Aptitude SECTION Quant Math Ability Mark(s) 1

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In a school, there are 500 students, of which 40% of students are girls and the remaining are boys. On a day if 15% of girls and 25% of boys are absent. How many students are present on that day?

☐ 425☐ 200☐ 395☐ 300

Previous Question

Next Question

QUESTION 9 GROUP: Aptitude SECTION: Logical Reasoning Mark(s): 1

Directions: Read the following information and answer the questions given below.

There are four friends A, B, C, D. One of them is a Cricketer and studies Chemistry and Biology. A and B play Football. Both Football players study Maths. D is a Boxer. One Football player also studies Physics. The Boxer studies Maths and Accounts. All the friends study two subjects each and play one game each.

Question: Who studies Accounts and plays Football?

☐ B alone

☐ A alone

☐ A or B

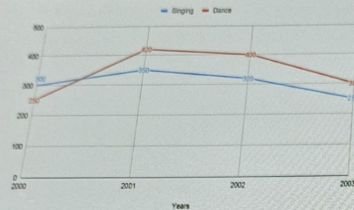
☐ D

Previous Question

Next Question

1

The line graph shows the number of students (male and female included) enrolled in singing and dance class from 2000 to 2003.



The table shows the number of students (male and female included) enrolled in the dance class from different age groups in different years.

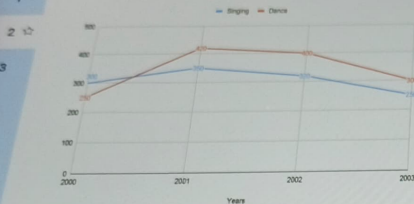
Years	Age groups		
	10-17 years	18-25 years	26-33 years
2000	46	60	55
2001	40	65	70
2002	55	60	64
2003	54	70	75

Question: What is the average number of students in the dance class whose age were either less than 10 years or more than 33 years in the given years?

☐ 164☐ 221☐ 101[Previous Question](#)[Next Question](#)

1

The line graph shows the number of students (male and female included) enrolled in singing and dance class from 2000 to 2003.



7

The table shows the number of students (male and female included) enrolled in the dance class from different age groups in different years.

8

9

Years	Age groups		
	10-17 years	18-25 years	26-33 years
2000	46	60	55
2001	40	65	70
2002	55	60	64
2003	54	70	75

11

Question:

12

If the age of 30% of the students from dance class in the year 2002 were more than 33 years, then how many students were there whose age was less than 10 years from dance class?

13

☐ 101

14

☐ 98

Previous Question

Next Question

QUESTION 10 GROUP: Aptitude SECTION: Logical Reasoning Mark(s): 1

2

3

Directions: Read the following information and answer the questions given below.

4

There are four friends A, B, C, D. One of them is a Cricketer and studies Chemistry and Biology. A and B play Football. Both Football players study Maths. D is a Boxer. One Football player also studies Physics. The Boxer studies Maths and Accounts. All the friends study two subjects each and play one game each.

5

Question: How many games are played and subjects studied by the four friends?

6

☐ 1 game and 4 subjects

7

☐ 3 games and 4 subjects

8

☐ 2 games and 3 subjects

9

☐ 3 games and 5 subjects

10 ☆

11

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Previous Question

Next Question

16

QUESTION 22 GROUP: English SECTION: Fill in the blanks Mark(s): 1

17

Choose the option that meaningfully completes the sentence.

18

After an ugly incident that took place five years ago, she _____ the grit to overcome it.

19

☐ be

20

☐ have

21

☐ had have

22 ☆

☐ had

23

24

25

Previous Question

Next Question

16

QUESTION: 24 GROUP: English SECTION: Reading Comprehension Mark(s): 1

17

Direction: Read the passage given below and answer the question that follows. There may be more than one question(s) based on the below passage.

18

We are surrounded by the sounds of the machinery that make our lives comfortable and convenient. The constant thrum of traffic and the thunder of jet engines overhead. But when we have to listen to these noises for too long or at the wrong time, they can inflict silent and stealthy damage. Increasing evidence shows this damage isn't just to our ears, but to our blood vessels and hearts. Nor is this just a problem for people who live near busy roads or under flight paths. New research suggests that noise pollution also causes harm in places such as hospitals.

19

20

Last week, the world's experts gathered in Japan to discuss the latest findings about noise and health. Perhaps most eye-catching was the study that linked noise pollution to your waist size. In a four-year project published last year, researchers from Karolinska University in Sweden found that the louder the traffic noise to which people in different parts of Stockholm were exposed, the greater the increase in their waist size - there was nearly a centimetre increase for every ten-decibel rise in the noise levels. This is like the difference between a whispered conversation and the noise level in an average house or personal office.

21

22

Last month, scientists from Karolinska University found an even more dramatic effect from plane noise. After tracking more than 5,000 people for ten years, they reported that the waistlines of those most exposed to plane noise increased on average by 6 cm. A similar study last year by the School of Public Health at Imperial College London found that being exposed to higher levels of aircraft noise around Heathrow raised the risk of admission to hospital for heart disease by 20 per cent. The effects of noise pollution are even felt by babies in the womb.

23

Question: The statement "We are surrounded by the sounds of the machinery that make our lives comfortable and convenient" implies that:

24 ☆

☐ it is good to be surrounded by machines always.

25

☐ machines make our lives comfortable by creating sound.☐ sound is the proof that machines are working.☐ most of the times we can observe the sound of machines around us.

Previous Question

Next Question

- 16 We are surrounded by the sounds of the machinery that make our lives comfortable and convenient. The constant thrum of traffic and the thunder of jet engines overhead. But when we have to listen to these noises for too long or at the wrong time, they can inflict silent and stealthy damage. Increasing evidence shows this damage isn't just to our ears, but to our blood vessels and hearts. Nor is this just a problem for people who live near busy roads or under flight paths. New research suggests that noise pollution also causes harm in places such as hospitals.
- 17
- 18 Last week, the world's experts gathered in Japan to discuss the latest findings about noise and health. Perhaps most eye-catching was the study that linked noise pollution to your waist size. In a four-year project published last year, researchers from Karolinska University in Sweden found that the louder the traffic noise to which people in different parts of Stockholm were exposed, the greater the increase in their waist size - there was nearly a centimetre increase for every ten-decibel rise in the noise levels. This is like the difference between a whispered conversation and the noise level in an average house or personal office.
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21 **Question:**

What are the likely health effects of noise pollution?

- 22 (I) It increases the risk of ear damage.
- 23 (II) It increases the risk of blood pressure.
- 24 (III) It increases the risk of obesity.
- (IV) It increases the risk of heart disease.

25 ☆

- ☒ Only (I), (III) and (IV)
- ☐ Only (III)
- ☐ All (I), (II), (III) and (IV)
- ☐ Only (I), (II) and (III)

Previous Question

Next Question

16 QUESTION 21 GROUP English SECTION Error Detection Mark(s) 1

17 Out of the four underlined phrases marked (A), (B), (C) and (D), choose the phrase which is wrong or inappropriate in the context of the sentence.

18 You will not (A) pass the (B) examination, till you (C) study hard. (D)19 ☐ C20 ☐ B21 ☆ ☐ D22 ☐ A

23

24

25

Previous Question

Next Question

What will be the result of the code given below?

32

```
int x = 1;  
for (x = 0; x < 2; x++)  
{
```

33

```
    switch(x)
```

34

```
    {
```

35

```
        case 0: x += x;
```

```
        case 1: x += x;
```

36

```
        case 5: x += x;
```

```
        default: x += x;
```

37 ☆

```
        break;
```

```
    }
```

38

```
    print(x);
```

```
}
```

39

40

☐ 0
☐ 6

41

☐ 0
☐ 16

42

43

☐ 0
☐ 2

44

☐ 0
☐ 8

45

[Previous Question](#)[Next Question](#)

32

33

What will be the output of the query given below?

34

```
SELECT COUNT (*) + COUNT (*) + COUNT (*) + 1;
```

35

☐ 0

36

☐ Throw error

37

☐ 4

38

☐ 3

39

40

41

42

43

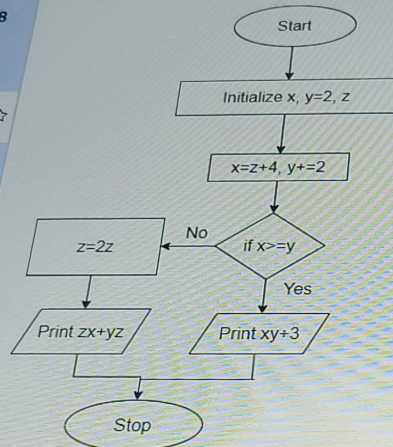
44 ☆



45

QUESTION 30 GROUP Technical SECTION: Programming Ability Mark(s): 1

Generate the output considering that the value of $z=1$ and choose the most appropriate option.



☐ 24

☐ 22

☐ 23

Previous Question

Next Question

32

33

What will be the result of the code given below?

34

```
int i = 12345;
```

```
int x, y = 0;
```

35

```
do
```

```
{
```

36

```
    x = i % 10;
```

```
    y = y * 10 + x;
```

37

```
    i = i / 10;
```

```
}
```

38

```
while (i > 1024);
```

```
print(y);
```

39 ☆

☐ 0

40

☐ 543

41

☐ 5

42

☐ 54

43

44

QUESTION 44 GROUP: Technical SECTION: SQL Mark(s): 1

What will be the output of the query given below?

```
SELECT COUNT (*) + COUNT (*) + COUNT (*) + 1;
```

☐ 0☒ Throw error☐ 4☐ 3[↶ Unanswer](#)

46 ☆

QUESTION 46

GROUP: Language Coding

SECTION: Code Nugget

Mark(s): 5

47

Write a program to find the number(s) which are contained in a string(str)

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings while reading the input or while printing, as these contribute to the standard output.

Input Format:

A single line of input contains a string(str).

Output Format:

A single line of output should display the list of all the numbers which are contained in a string separated by a single white space.

Sample Input

12john575859

Sample Output:

1 2 5 7 5 8 5 9

Explanation:

The given string is 12john575859

So after combining every word of the string we get "1 2 5 7 5 8 5 9"

Current Code

Previous Versions

C++

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 void printResult(char *s)           // s is the input string.
5 {
6
7     // WRITE YOUR CODE HERE
8     vector<char> v;
9     for(int i=0;i<s.size();i++){
10         if
11     }
12 }
13
14 int main()
15 {
16     char *str;
17     Str = (char *)malloc(sizeof(char)*100);
18     cin >> Str;
19     printResult(Str);
20     return 0;
21 }
```

☐ Test against custom input

Previous Question

Next Question

46

QUESTION: 47 GROUP Language Coding SECTION Coding 1 Mark(s) 10

47

Write a program that takes a list of words L as input and sorts the list of words in ascending alphabetical order. From the sorted list, identify the words that has the given character C in it and print the original words along with their positions in the sorted list if it has the specified character in it.

Read the input from STDIN and print the output to STDOUT. Do not print arbitrary strings anywhere in the program, as these contribute to the standard output, and test cases will fail.

Constraints:

- i) The string consists of lowercase letters only.
- ii) if none of the word has the given character print 0 as output.

Input Format:

The first line of input contains the string L , the given string in the lowercase.
The second line of input contains the character, C , the character to be searched.

Output Format:

The output should print the list of words that has the character C in it and its position. Both should be separated by a single white space and each word should be in a separate line. words must be displayed in the order of sorted list.

Sample Input 1:

this is for testing the data
t

Sample Output 1:

data 1
testing 4
the 5
this 6

Explanation 1:

Here, from Sample Input 1 the given list, L is "this is for testing the data"
After sorting all the words of the list L , we get the sorted list as "data for is

Current Code

Previous Versions

C++

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 void sortWordsAndSearchChar(char inputStr[], char searchCh) { // inputStr and searchCh are
    variable
5
6     // WRITE YOUR CODE HERE.
7
8 }
9
10 int main() {
11     char inputString[100], searchChar;
12     cin.getline(inputString, sizeof(inputString));
13     cin >> searchChar;
14     sortWordsAndSearchChar(inputString, searchChar);
15     return 0;
16 }
17
```

☐ Test against custom input

Previous Question

Next Question

46

QUESTION: 47 GROUP: Language Coding SECTION: Coding 1 Mark(s): 10

47

the 5
this 6**Explanation 1:**

Here, from Sample Input 1 the given list, L is "this is for testing the data". After sorting all the words of the list L , we got the sorted list as "data for is testing the this".

Therefore, now we can clearly see that the character "t" is in words; "data, testing, the, this" which are the 1st, 4th, 5th, 6th words in the sorted list. So, it will print

data 1
testing 4
the 5
this 6
as output.

Sample Input 2:

planet earth is our home
e

Sample Output 2:

earth 1
home 2
planet 5

Explanation 2:

Here, from Sample Input 2 the given list ' L ' is "planet earth is our home". After sorting all the words of the list L , we got the sorted list as "earth home is our planet".

So, it will print

earth 1
home 2
planet 5

as output because only these words are having 'e' in them.

Current Code

Previous Versions

C++

```

16- for(int i{}; i < words.size(); i++){
17-     if(words[i].find(searchChar) != string::npos){
18-         cout << words[i] << " " << (i+1) << endl;
19-         found = true;
20-     }
21- }
22- if(!found) cout << 0 << endl;
23- }
24-
25- int main() {
26-     char inputString[100], searchChar;
27-     cin.getline(inputString, sizeof(inputString));
28-     cin >> searchChar;
29-     sortWordsAndSearchChar(inputString, searchChar);
30-     return 0;
31- }
32-

```

☐ Test against custom input
Test Cases Results

1	Sample TC	Pass
2	Sample TC	Pass
3		Fail
4		Fail
5		Pass
6		Fail
7		Fail

Result : Test case passed

Time (sec)

0.00422

Memory (KiB)

3584

Compiler Message

Compilation Successful.

Warning : solution.cpp: In function

solution.cpp:16:19: warning: compar

16 | for(int i{}; i < words.size(); i++){

Previous Question

Next Question

46

QUESTION: 47 GROUP: Language Coding SECTION: Coding 1 Mark(s): 10

47 ☆

Write a program that takes a list of words L as input and sorts the list of words in ascending alphabetical order. From the sorted list, identify the words that has the given character C in it and print the original words along with their positions in the sorted list if it has the specified character in it.

Read the input from STDIN and print the output to STDOUT. Do not print arbitrary strings anywhere in the program, as these contribute to the standard output, and test cases will fail.

Constraints:

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Input Format:

The first line of input contains the string L , the given string in the lowercase.
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Output Format:

The output should print the list of words that has the character C in it and its position. Both should be separated by a single white space and each word should be in a separate line. Words must be displayed in the order of sorted list.

Sample Input 1:

this is for testing the data
t

Sample Output 1:

data 1
testing 4
the 5
this 6

Explanation 1:

Here, from Sample Input 1 the given list, L is "this is for testing the data". After sorting all the words of the list L , we got the sorted list as "data for is testing the this". Therefore, now we can clearly see that the character "t" is in words: "data, testing, the, this" which are the 1st, 4th, 5th, 6th words in the sorted list.

Current Code

Previous Versions

C++

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 void sortWordsAndSearchChar(char inputStr[], char searchCh) {
5     inputStr and searchCh is a string variable
6     // WRITE YOUR CODE HERE.
7     vector<string> words;
8     stringstream ss(inputStr);
9     string word;
10
11     while(ss >> word){
12         words.push_back(word);
13     }
14     if(words.empty()) return;
15     sort(words.begin(), words.end());
16     bool found = false;
17     for(int i=0; i < (int)words.size(); i++){
```

☐ Test against custom input

Run Tests

Submit

Test Cases Results1 Samp
le TCPas
s2 Samp
le TCPas
s

3

Fail

4

Fail

5

Pass

6

Fail

Result : Test case passed

Time (sec)

0.00424

Memory (KiB)

3584

Compiler Message

Compilation Successful.

Input

this is for testing the data

Previous Question

Next Question

oding 1 Mark(s): 10

sorts the list of words
identify the words that has
long with their positions

T. Do not print
ute to the standard

out.

the lowercase.
ter to be searched.

C in it and its
each word
der of sorted list.

ata"
feria

Current Code

Previous Versions

C++

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 void sortWordsAndSearchChar(char inputStr[], char searchCh) { // inputStr and searchCh is an string
5     variable
6     // WRITE YOUR CODE HERE.
7
8     vector<string> words;
9     stringstream ss(inputStr);
10    string word;
11
12    while(ss >> word){
13        words.push_back(word);
14    }
15    if(words.empty()) return;
16    sort(words.begin(), words.end());
17    bool found = false;
18    for(int i=0; i < (int)words.size(); i++){
19        if(words[i].find(searchCh) != string::npos){
20            cout << words[i] << " " << i+1 << "\n";
21            found = true;
22        }
23    }
24    if(!found) cout << 0;
25 }
26
27 int main() {
28     char inputString[100], searchChar;
29     cin.getline(inputString, sizeof(inputString));
30     cin >> searchChar;
31     sortWordsAndSearchChar(inputString, searchChar);
32     return 0;
33 }
34
```

☐ Test against custom input

Run Tests

Submit & Continue Coding

Previous Question

Next Question

End Test